Visual Differential-Spatially Projected Transformer for Efficient Hyperspectral Images Super-Resolution

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***Abstract*—Single hyperspectral image (HSI) super-resolution aims to restore high-resolution HSIs from a single low-resolution HSI. Existing methods often struggle with inefficient spatial attention mechanisms and inadequate modeling of cross-spectral dependencies, which hinder the simultaneous achievement of high restoration accuracy and computational efficiency. To address these limitations, we propose an efficient yet powerful architecture named the visual differential-spatially projected transformer (VDSPT). Instead of conventional spatial attention, VDSPT introduces visual differential–spatial attention (VDSA), which computes two distinct attention maps and subtracts one from the other to suppress overemphasized regions. This differential strategy enables more balanced and effective extraction of spatial features. In addition, we design spatially projected spectral attention (SPSA), which projects features along the spatial dimension before computing spectral attention while preserving the natural ordering of spectral channels. This leads to more robust and reliable modeling of interchannel spectral relationships. By stacking only a few VDSA and SPSA modules, VDSPT achieves the state-of-the-art (SOTA) superresolution performance with remarkable computational efficiency. Extensive experiments on both natural and remote sensing hyperspectral datasets demonstrate that VDSPT consistently outperforms existing methods in terms of quantitative metrics and visual quality. Our codes are available at https://github.com/bbdswbf/VDSPT**

***Index Terms*—Efficient vision transformer, hyperspectral image superresolution (HSI-SR), spatially projected spectral attention (SPSA), visual differential–spatial attention (VDSA).**

I. Introduction

HYPERSPECTRAL imaging is an advanced image acquisition technique capable of continuously capturing numerous narrowband spectral information across a broad electromagnetic spectrum [1], [2], [3]. Each pixel in hyperspectral images (HSIs) contains rich spectral characteristics, making it valuable for applications in agriculture [4], environmental monitoring [5], and medical diagnostics [6], [7]. However, due to the physical constraints of imaging sensors, HSIs often suffer from low spatial resolution, which hinders their effectiveness in practical scenarios [8], [9]. To address this limitation, HSI superresolution (HSI-SR) emerges as a critical research area, aiming to enhance the spatial quality of HSIs through computational methods, thereby enabling more accurate and reliable downstream analysis.

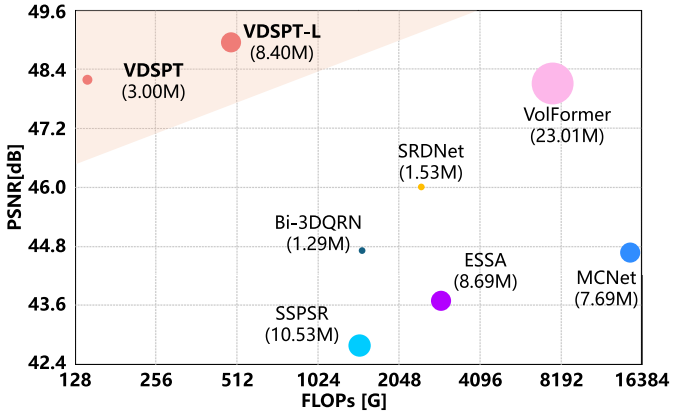


Fig. 1. Comparison of PSNR, model parameter size (Params), and FLOPs between our VDSPT and SOTA single HSI-SR methods on ARAD_1k dataset at $\times 2$ scale. VDSPT achieves the best performance in both restoration quality and computational efficiency.

HSI-SR techniques can be categorized into fusion-based approaches [10], [11], [12], [13], [14], [15], [16], [17] and single HSI-SR [18], [19], [20], [21], [22]. While the former theoretically benefits from multisource data complementarity, it faces practical limitations, as acquiring spatiotemporally consistent auxiliary data is often infeasible in applications such as remote sensing or medical imaging. On the contrary, single HSI-SR solely utilizes the intrinsic information entropy of low-resolution HSI (LR HSI) to restore high-resolution HSI (HR HSI), providing critical advantages in resource-constrained environments.

In recent years, deep-learning-based approaches [18], [19], [20], [21], [23], [24], [25], [26] have dominated the field of single HSI-SR. Among them, methods that leverage attention mechanisms [21], [25], [26] to model long-range spatial and spectral dependencies have achieved remarkable super-resolution performance. Even recent representative works, including efficient lightweight Transformer frameworks [27] and unified HSI restoration frameworks [28], [29], still follow the conventional attention design paradigm. However, due to inefficient spatial feature extraction and unstable spectral feature modeling, these methods often require stacking a large number of modules to attain high-quality restoration, resulting in substantial computational overhead. In particular, existing single HSI-SR methods suffer from two critical issues.

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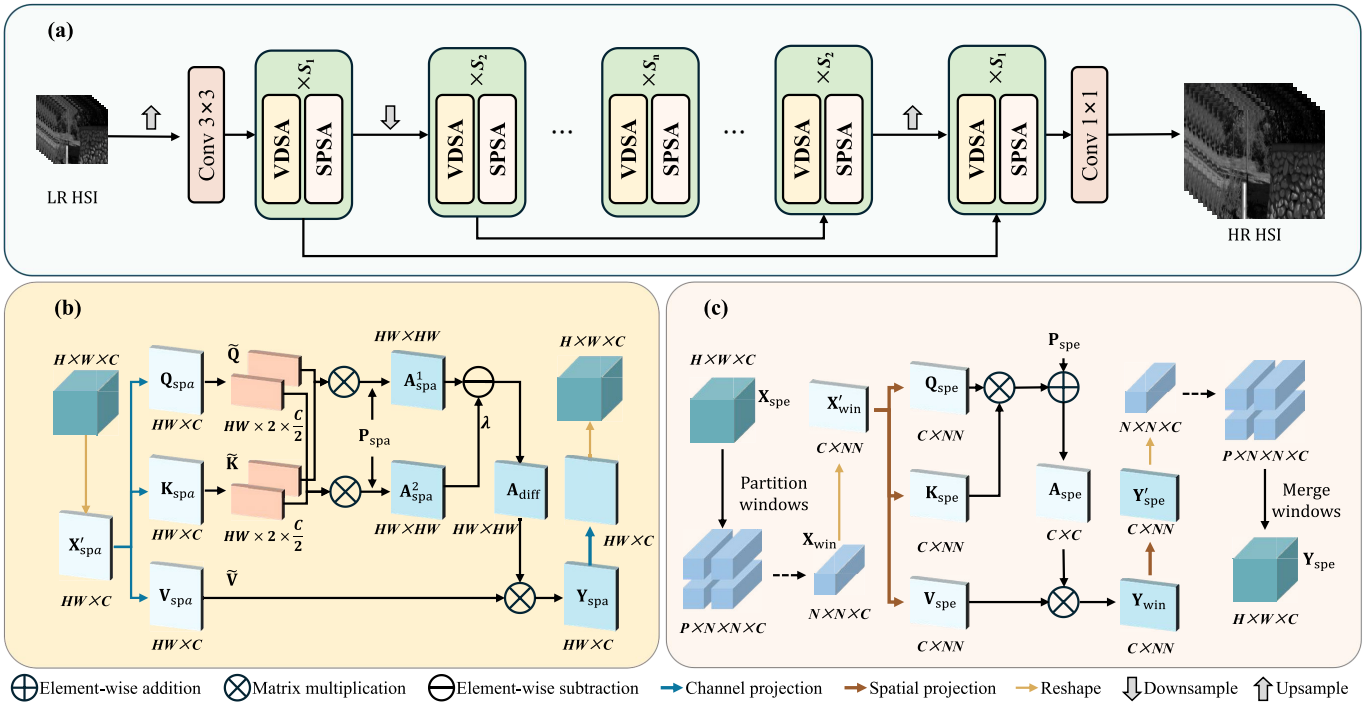


Fig. 2. (a) Overall framework of visual differential-spatially projected transformer. (b) Pipeline of visual differential–spatial attention. (c) Pipeline of spatially projected spectral attention.

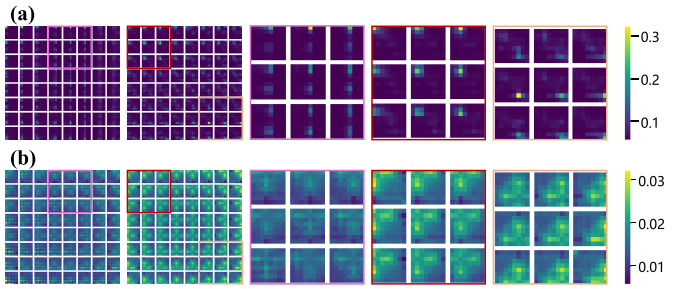


Fig. 3. Visual comparison between conventional visual attention and VDSA. For (a) and (b) two images on the left are attention visualization maps, and the three images on the right are magnified views of the areas outlined by the corresponding color boxes. Each attention visualization map represents the attention weights of each pixel to all pixels within an 8 × 8 pixel block. For instance, in a subplot, the first block (of shape 8 × 8) at the top-left corner indicates the attention weights of the top-left pixel to all other pixels. (a) Conventional visual attention tends to assign extremely high weights (over 30%) to itself or its immediate neighbors, while (b) VDSA distributes attention more evenly across the global context, with most individual attention weights staying below 5%.

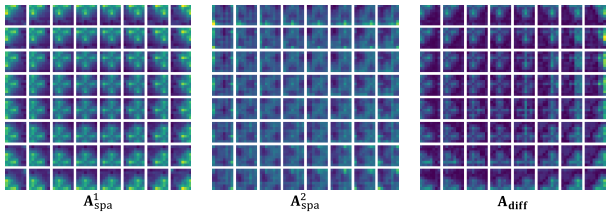


Fig. 4. Attention map visualization of VDSA. (Left) to (Right): $\mathbf{A}^1_{\text{spa}}$ (signal branch), $\mathbf{A}^2_{\text{spa}}$ (calibration branch), and $\mathbf{A}_{\text{diff}}$ (final differential attention).

1) Imbalanced Spatial Attention Distributions Lead to Inefficient Spatial Feature Extraction: Conventional spatial attention mechanisms tend to assign excessively high weights to only a small subset of pixels, particularly those in local neighborhoods, whereas just one pixel can account for over 30% of the total attention weight. This bias stems from the fact that a pixel is inherently most similar to itself. In superresolution tasks, however, local information is inherently limited, and restoring fine details relies heavily on global contextual cues. Overemphasizing self-similarity and immediate neighbors thus significantly reduces the efficiency of global spatial feature extraction.

2) Channel Projection Before Spectral Attention Computation Leads to Unstable Spectral Feature Representation: Most existing methods apply independent linear transformations across channels to generate query (Q), key (K), and value (V) vectors before computing spectral attention. Because these transformations use different parameters, the channel ordering in Q, K, and V vectors becomes inconsistent. Since spectral attention is intended to capture intrinsic relationships among the original input channels, this misalignment in channel correspondence introduces ambiguity and degrades the reliability of spectral attention estimation. To compensate for these inefficient attention mechanisms and suboptimal feature representations, current approaches resort to stacking numerous modules, making it difficult to simultaneously achieve high superresolution performance and low computational cost.

In this article, we propose the visual differential-spatially projected transformer (VDSPT), a lightweight yet powerful framework for single HSI-SR, comprising two core components, visual differential spatial attention (VDSA) and spatially projected spectral attention (SPSA). In particular, VDSA splits the Q and K vectors along the channel dimension into two complementary groups, computes two separate attention maps, and performs elementwise subtraction to obtain a differential

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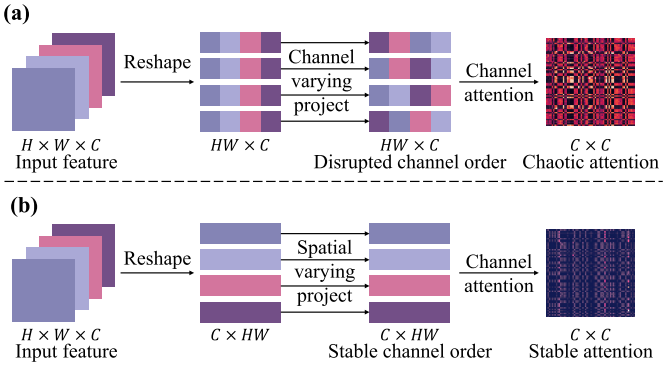


Fig. 5. Comparison between conventional spectral attention and SPSA. The different colors of the features represent the corresponding initial channel numbers. (a) Conventional spectral attention mechanisms perform channel-varying projection, disrupting the original channel order, which further leads to chaotic attention estimation when calculating channel attention. (b) SPSA adopts spatial varying projection, preserving the ordered relationship among the channels and enabling more stable channel attention computation.

attention output. This operation effectively calibrates the over-concentrated local weight bias caused by Softmax constraints, such as those at self-positions or nearby pixels, thereby enhancing contextual awareness of global features and improving spatial feature extraction efficiency. For spectral modeling, unlike conventional approaches that apply channelwise projections before spectral attention computation (which often disrupts spectral coherence), SPSA performs projection along the spatial dimension. This ensures consistent channelwise distributions across Q, K, and V vectors, matching those before projection, thereby leading to more stable spectral attention computation. Moreover, by conducting spatial projection and spectral attention within local windows, SPSA circumvents the issue of incompatible input sizes, enabling the use of fixed-size projection weights. Furthermore, we cascade VDSA and SPSA in a decoupled manner to jointly extract spatial and spectral features. Thanks to VDSA's efficient spatial modeling and SPSA's robust spectral representation, VDSPT achieves high-performance superresolution with only a small number of cascaded modules and minimal computational cost. As shown in Fig. 1, VDSPT achieves the state-of-the-art (SOTA) results in both quality and efficiency.

Our main contributions are summarized as follows.

1) We introduce VDSA that enhances global perception by suppressing excessive focus on dominant local regions through a differential attention mechanism.
2) We design SPSA, which ensures stable spectral modeling by avoiding disruptive channel projections and instead leveraging spatially varying feature aggregation.
3) We present VDSPT, a lightweight yet powerful framework that achieves SOTA performance on multiple HSI-SR benchmarks while maintaining high efficiency.

II. Related Work

In this section, we review the related work on single HSI-SR and efficient visual transformers and highlight the key challenges associated with each.

A. Single HSI-SR

Single HSI-SR can be categorized into conventional, convolution-based, and attention-based methods. Conventional methods use strategies such as iterative back projection [30] and sparse representation [31] for superresolution, relying on handcrafted priors to model the underlying structure of HSI. While conceptually intuitive, these approaches often require careful parameter tuning and exhibit limited performance in complex scenes due to their inflexible modeling assumptions.

With the advent of deep learning, 3-D convolutional networks [18], [23] have been widely adopted to capture spatial–spectral correlations in a data-driven manner. These networks apply volumetric filters across both spatial and spectral dimensions, enabling joint feature learning. However, their cubic computational complexity with respect to spatial resolution and spectral band count poses a significant bottleneck for practical deployment. As a result, hybrid 3-D and 2-D convolution methods [24], [32] come into being, which decouple spatial and spectral processing to achieve a more favorable balance between performance and efficiency.

Despite these improvements, convolutional methods inherently suffer from limited local receptive fields, which restricts their ability to model long-range spatial dependencies and global contextual patterns that are crucial for restoring fine textures and structural details. This limitation has motivated the adoption of attention mechanisms, which offer a global receptive field and adaptive feature recalibration. In this context, SSPSR [19] uses a channel attention mechanism for spectral feature extraction, emphasizing informative bands while suppressing less relevant ones. ESSA [25] further improves efficiency by introducing a kernelizable attention mechanism that reduces computational overhead. SRDNet [26] employs a spatial attention pyramid to capture self-similarity across different scales, enhancing the restoration of repetitive structures. Meanwhile, EigenSR [20] adopts a spatial–spectral decoupled design by leveraging a pretrained RGB transformer for spatial feature extraction, combined with spectral decomposition. More recently, VolFormer [21] decomposes 3-D spatial–spectral attention into dual similarity matrices, effectively enhancing interactions across both spatial and spectral dimensions through factorized modeling.

Despite their excellent performance, attention-based methods still face significant challenges. The distribution of spatial attention is often highly uneven, with excessive focus on trivial or self-attended regions, leading to inefficient context aggregation. Moreover, the channelwise feature projection commonly used in spectral attention mechanisms disrupts the natural ordering of spectral bands, resulting in disordered and unstable estimation of spectral dependencies. To address these issues, our VDSPT mitigates spatial attention overconcentration and preserves spectral consistency through VDSA and SPSA.

B. Efficient Visual Transformers

The transformer architecture [33], originally developed for natural language processing and later adapted to computer vision [34], demonstrates strong performance in image restoration tasks such as superresolution, denoising, and deblurring.

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However, its computational complexity grows quadratically with spatial resolution, leading to high computational costs in dense prediction tasks like HSI-SR. To address this challenge, extensive research focuses on developing efficient transformers [35]. Representative strategies include window-based attention mechanisms [36], which limit self-attention computation to local windows to achieve linear complexity. Other approaches involve module-level optimizations through feed-forward network refinement and simplified multihead attention [37]. Additional techniques include memory-efficient training methods such as sparse activation [38].

Nevertheless, these methods perform poorly on single HSI-SR because they overlook the intrinsic properties of hyperspectral data, particularly its rich spectral information. On the contrary, our joint use of VDSA and SPSA simultaneously leverages the spatial–spectral characteristics of HSI to achieve efficient and effective single HSI-SR.

III. Methodology

In this section, we first formally define the single HSI-SR problem, then present the overall framework of the proposed VDSPT, followed by a detailed explanation of the core ideas and implementations of its two key modules, VDSA and SPSA.

A. Problem Formulation

Single HSI-SR aims to restore HR HSIs $\mathbf{Y} \in \mathbb{R}^{H\times W\times C}$ from their corresponding LR HSIs $\mathbf{Z} \in \mathbb{R}^{h\times w\times C}$, where $H = s \cdot h$, $W = s \cdot w$, and s denotes the spatial upsampling scale factor. Both $\mathbf{Z}$ and $\mathbf{Y}$ share the same number of spectral bands C, but differ in spatial resolution.

The degradation process from HR HSI to LR HSI is commonly modeled as

$$\mathbf{Z} = \mathcal{D}_s\left(\mathcal{B}\left(\mathbf{Y}\right)\right) \tag{1}$$

where $\mathcal{B}(\cdot)$ represents a spatial blurring operator (e.g., Gaussian blur) and $\mathcal{D}_s(\cdot)$ denotes spatial downsampling by a factor of s. The goal of single HSI-SR is to learn a mapping function $f_\theta(\cdot)$ parameterized by θ, such that

$$\hat{\mathbf{Y}} = f_\theta\left(\mathbf{Z}\right) \tag{2}$$

where $\hat{\mathbf{Y}}$ is the restored HR HSI, and the model parameters θ are optimized to minimize the restoration error under both spatial and spectral fidelity constraints.

Restoring an HR HSI from an LR HSI, in which high-frequency details are degraded or lost, remains highly challenging, as it requires effective joint modeling and exploitation of spatial and spectral features. Most existing methods address this by stacking deep networks. In this work, we introduce VDSA for efficient spatial feature extraction and SPSA for robust spectral feature representation, achieving high performance and computationally efficient HSI-SR.

B. Overall Framework

As shown in Fig. 2(a), for the input LR HSI, we first upsample it to the same spatial resolution as the HR HSI using bilinear interpolation. A 3×3 convolution is then applied to extract shallow features. Subsequently, the features are processed through multiple encoder–decoder stages. In each encoder stage, features are extracted, followed by downsampling. In the corresponding decoder stage, the network takes as input both the encoded features from the matching encoder stage and the upsampled features from the previous decoder stage, and produces the decoded features for the current stage. Finally, a residual block fuses the shallow features with the enhanced features from the decoder to generate the restored image. Notably, both the encoder and decoder stages consist of cascaded VDSA and SPSA modules to achieve efficient and stable spatial and spectral feature extraction.

C. Visual Differential–Spatial Attention

We conduct a visual analysis of conventional vanilla self-attention, as shown in Fig. 3(a). Constrained by the nonnegative and sum-to-one simplex rule of the Softmax function, conventional attention falls into a zero-sum weight game in which it tends to assign disproportionately high weights to the query position itself and its immediate neighborhood while squeezing the weights of nearly all global regions to near zero. This highly localized allocation makes weighted feature aggregation nearly equivalent to the identity mapping of input features, and wastes most attention capacity on redundant self-correlated signals. It thus fails to capture global contextual information and model long-range non-local self-similarity, which serves as a core prior for high-quality HSI-SR. This limitation directly restricts its ability to exploit spatial correlations across the feature map and leads to redundant features and weak contextual modeling.

To mitigate this overconcentration of attention on self-positions, we propose the VDSA. This module calibrates the redundant local weight bias via elementwise subtraction between two independently learned attention maps, with the core operation formulated in (10). The two maps are denoted as the signal branch $\mathbf{A}^1_{\text{spa}}$ and the calibration branch $\mathbf{A}^2_{\text{spa}}$,respectively. The signal branch captures task-relevant spatial correlations critical for SR reconstruction, including local texture features, edge information, and long-range nonlocal dependencies. The calibration branch learns a global baseline matching the inherent redundant bias of vanilla attention, and spontaneously converges to a nearly uniform global distribution without capturing task-relevant image content. The subtraction operation reduces spurious overconcentrated weights caused by the Softmax zero-sum constraint, while effectively retaining valid task-relevant signals for high-fidelity SR. Notably, this functional separation between the two branches is a natural optimal convergence outcome under the end-to-end SR reconstruction objective, and requires no additional explicit constraints such as orthogonality regularization or auxiliary losses. Meanwhile, the subtraction-based differential operation effectively relaxes the nonnegative and sum-to-one constraints of Softmax, an effect that is difficult to achieve with common linear fusion strategies.

Fig. 3(b) presents the overall attention weight distribution of VDSA. As shown, VDSA achieves a more balanced,

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contextually informed attention allocation across the global feature space and effectively alleviates the extreme local overconcentration observed in conventional attention. Notably, it still assigns higher yet moderate weights to spatially proximal regions and preserves relative positional information and essential local texture details for SR reconstruction.

To further verify the dual-branch functional separation and intended differential calibration logic, we supplement independent attention map visualization of VDSA shown in Fig. 4. The visualization is arranged from left to right as $\mathbf{A}^1_{\text{spa}}$ (signal branch), $\mathbf{A}^2_{\text{spa}}$ (calibration branch), and $\mathbf{A}_{\text{diff}}$ (final differential attention). The results directly validate our core design. The left panel presents a structured, content-aware distribution with reasonable focus on edge and texture regions, and it retains SR-critical spatial features well. The middle panel exhibits a nearly uniform global distribution without content-related response, and aligns well with our designed bias cancellation baseline. The right panel well retains the effective signal distribution of the signal branch while calibrating redundant global bias, and intuitively verifies the effectiveness of our differential mechanism.

In particular, given an input feature $\mathbf{X}_{\text{spa}} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent the height, width, and channel dimensions, respectively, the process begins by flattening the spatial dimension into $\mathbf{X}'_{\text{spa}} \in \mathbb{R}^{HW\times C}$. Subsequently, we perform channel varying projection. Unlike previous works [26], [36], we incorporate residual connections after projection to ensure more stable results

$$\mathbf{Q}_{\text{spa}} = \mathbf{W}^Q_{\text{spa}}\mathbf{X}'_{\text{spa}} + \mathbf{X}'_{\text{spa}} \tag{3}$$

$$\mathbf{K}_{\text{spa}} = \mathbf{W}^K_{\text{spa}}\mathbf{X}'_{\text{spa}} + \mathbf{X}'_{\text{spa}} \tag{4}$$

$$\mathbf{V}_{\text{spa}} = \mathbf{W}^V_{\text{spa}}\mathbf{X}'_{\text{spa}} + \mathbf{X}'_{\text{spa}} \tag{5}$$

where $\mathbf{Q}_{\text{spa}}, \mathbf{K}_{\text{spa}}, \mathbf{V}_{\text{spa}} \in \mathbb{R}^{HW\times C}$ denotes the Q, K, and V vectors, and $\mathbf{W}^Q_{\text{spa}}, \mathbf{W}^K_{\text{spa}}, \mathbf{W}^V_{\text{spa}} \in \mathbb{R}^{C\times C}$ are the projection weights. Next, we split the vectors along the channel dimension to form multihead attention vectors $\widetilde{\mathbf{Q}}, \widetilde{\mathbf{K}}, \widetilde{\mathbf{V}}$

$$\widetilde{\mathbf{Q}} = \text{reshape}\left(\mathbf{Q}_{\text{spa}}, (2H_{\text{eff}}, HW, D)\right) \tag{6}$$

$$\widetilde{\mathbf{K}} = \text{reshape}\left(\mathbf{K}_{\text{spa}}, (2H_{\text{eff}}, HW, D)\right) \tag{7}$$

$$\widetilde{\mathbf{V}} = \text{reshape}\left(\mathbf{V}_{\text{spa}}, (H_{\text{eff}}, HW, 2D)\right) \tag{8}$$

where H_{eff} is the number of attention heads for $\widetilde{\mathbf{V}}$, while $\widetilde{\mathbf{Q}}$ and $\widetilde{\mathbf{K}}$ have $2H_{\text{eff}}$ heads (Fig. 2(b) corresponds to the case where $H_{\text{eff}} = 1$). D denotes the channel dimension per head for $\widetilde{\mathbf{Q}}$ and $\widetilde{\mathbf{K}}$, with $C = 2H_{\text{eff}}D$. The attention matrix is then computed as

$$\mathbf{A}_{\text{spa}} = \text{softmax}\left(\frac{\text{norm}\left(\widetilde{\mathbf{Q}}\right)\cdot \text{norm}\left(\widetilde{\mathbf{K}}\right)^{\top}}{\mathbf{d}_{\text{spa}}} + \mathbf{P}_{\text{spa}}\right) \tag{9}$$

where $\mathbf{A}_{\text{spa}} \in \mathbb{R}^{2H_{\text{eff}}\times HW\times HW}$ is the attention matrix, $\mathbf{d}_{\text{spa}} \in \mathbb{R}^{2H_{\text{eff}}}$ is a scaling factor and $\mathbf{P}_{\text{spa}} \in \mathbb{R}^{2H_{\text{eff}}\times HW\times HW}$ denotes the position embedding. We then split $\mathbf{A}_{\text{spa}}$ along the head dimension into two components: $\mathbf{A}^1_{\text{spa}}, \mathbf{A}^2_{\text{spa}} \in \mathbb{R}^{H_{\text{eff}}\times HW\times HW}$. The differential attention is computed by subtracting them

$$\mathbf{A}_{\text{diff}} = \mathbf{A}^1_{\text{spa}} - \boldsymbol{\lambda}\mathbf{A}^2_{\text{spa}} \tag{10}$$

where $\mathbf{A}_{\text{diff}} \in \mathbb{R}^{H_{\text{eff}}\times HW\times HW}$ is the differential attention matrix, $\boldsymbol{\lambda} \in \mathbb{R}^{2H_{\text{eff}}}$ is a learnable parameter. The attention output is then calculated as

$$\mathbf{Y}_{\text{spa}} = \mathbf{A}_{\text{diff}}\widetilde{\mathbf{V}} \tag{11}$$

where $\mathbf{Y}_{\text{spa}} \in \mathbb{R}^{H_{\text{eff}}\times HW\times 2D}$ is the output. Finally, $\mathbf{Y}_{\text{spa}}$ is projected along the channel dimension and reshaped to the same shape as $\mathbf{X}_{\text{spa}}$ to produce the final output.

D. Spatially Projected Spectral Attention

The conventional spectral attention mechanism first computes the Q, K, and V vectors through channelwise transformation projections and applies the same global spectral attention to all pixels, as shown in Fig. 5(a). To generate distinct Q, K, and V vectors, different projection weights are required. However, applying separate projections along the channel dimension disrupts the original channel ordering among Q, K, and V vectors. Computing spectral attention under such misaligned channel configurations results in a disordered attention matrix. As illustrated in Fig. 5(a), this channelwise projection in the conventional approach scrambles the channel order, which in turn leads to chaotic spectral attention patterns where attention weights are almost exclusively either extremely high (red) or extremely low (black). Moreover, although different materials exhibit distinct spectral curve distributions, this mechanism enforces a single global spectral attention map across all pixels, compelling the model to capture all spectral dependencies through one shared attention matrix. This inevitably causes spectral attention blurring.

To address these issues, we propose the SPSA, which enables stable and fine-grained spectral attention estimation by performing projections along the spatial dimension combined with a window-based strategy. Given an input tensor, we first partition it into multiple nonoverlapping windows. Within each window, the Q, K, and V vectors are obtained by projecting features along the spatial dimension, thereby preserving the original channel ordering consistent with that of the input. Spectral attention is then computed within each window based on these aligned representations, yielding more stable attention responses. Because the window size is fixed, the same set of projection weights can be applied uniformly across windows, regardless of the overall input spatial dimension. As shown in Fig. 5(b), the SPSA mechanism produces a well-structured attention matrix: each spectral channel exhibits differentiated and nuanced attention weights toward other channels, in stark contrast to the extreme and uninformative responses observed in conventional spectral attention. Furthermore, computing spectral attention independently within each window enables the modeling of fine-grained spectral dependencies specific to local image structures.

In particular, given an input feature $\mathbf{X}_{\text{spe}} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C are the height, width, and channel dimensions, respectively, we first partition it into windows

$$\bigcup_{i=1}^{\lceil H/N\rceil}\bigcup_{j=1}^{\lceil W/N\rceil}\mathbf{X}^{ij}_{\text{spe}} = \mathbf{X}_{\text{spe}} \tag{12}$$

where i and j are the indices along the height and width dimensions of windows, N is the window size, and $\mathbf{X}^{ij}_{\text{spe}} \in \mathbb{R}^{N\times N\times C}$

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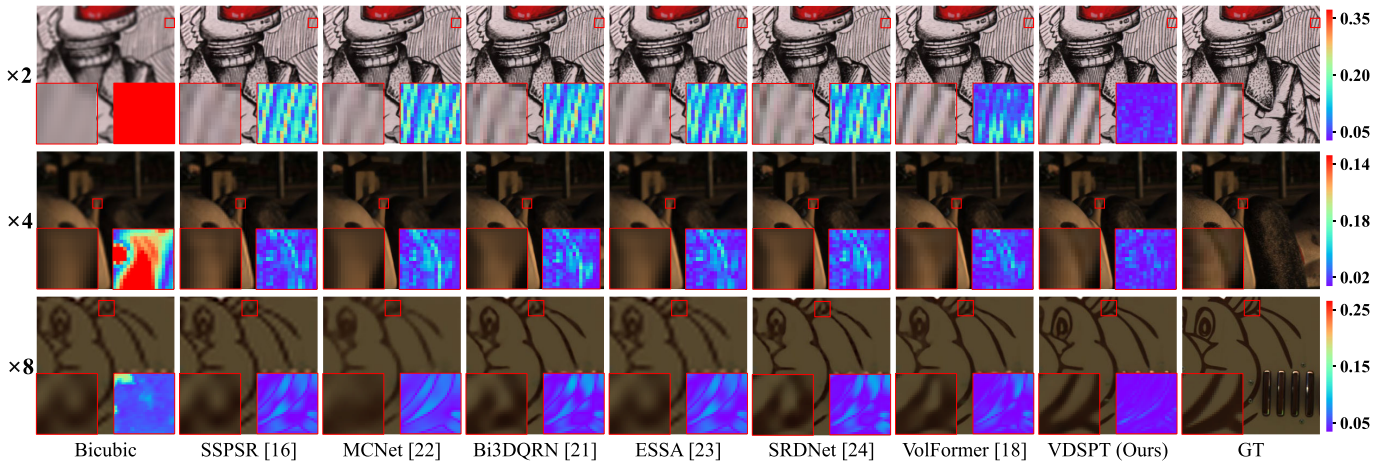


Fig. 6. Visual comparison on ARAD_1K [42] at ×2, ×4, ×8 scales. We use band numbers (27, 14, 6) to generate the pseudocolor images. (Bottom-left) for each result, selected details are highlighted with red boxes, and (bottom-right) the corresponding superresolution results and error maps are enlarged.

is the windowed feature. For clarity of exposition, we use $\mathbf{X}_{\text{win}}$ to denote $\mathbf{X}_{\text{spe}}^{ij}$.

Then, $\mathbf{X}_{\text{win}}$ is flattened along the spatial dimension

$$\mathbf{X}'_{\text{win}} = \text{vec}\left(\mathbf{X}_{\text{win}}\right) \tag{13}$$

where vec(·) denotes the operation of flattening along the spatial dimension followed by a spatial-channel dimension exchange, $\mathbf{X}'_{\text{win}} \in \mathbb{R}^{C\times N^2}$ is the flattened feature.

After that, we perform spatial projection

$$\mathbf{Q}_{\text{spe}} = \mathbf{W}_{\text{spe}}^{Q}\mathbf{X}'_{\text{win}}, \mathbf{K}_{\text{spe}} = \mathbf{W}_{\text{spe}}^{K}\mathbf{X}'_{\text{win}}, \mathbf{V}_{\text{spe}} = \mathbf{W}_{\text{spe}}^{V}\mathbf{X}'_{\text{win}} \tag{14}$$

where $\mathbf{Q}_{\text{spe}}, \mathbf{K}_{\text{spe}}, \mathbf{V}_{\text{spe}} \in \mathbb{R}^{C\times N^2}$ are the Q, K, and V vectors, and $\mathbf{W}_{\text{spe}}^{Q}, \mathbf{W}_{\text{spe}}^{k}, \mathbf{W}_{\text{spe}}^{v} \in \mathbb{R}^{N^2\times N^2}$ are the corresponding spatial projection weights.

Subsequently, the spectral attention is computed

$$\mathbf{A}_{\text{spe}} = \text{softmax}\left(\frac{\text{norm}\left(\mathbf{Q}_{\text{spe}}\right)\text{norm}\left(\mathbf{K}_{\text{spe}}^{\top}\right)}{\mathbf{d}_{\text{spe}}} + \mathbf{P}_{\text{spe}}\right) \tag{15}$$

where $\mathbf{A}_{\text{spe}} \in \mathbb{R}^{C\times C}$ is the spectral attention matrix, norm is the normalization along the spatial dimension, $\mathbf{P}_{\text{spe}} \in \mathbb{R}^{C\times C}$ is the spectral position embedding, $\mathbf{d}_{\text{spe}}$ is the scale factor.

Following this, output features are generated

$$\mathbf{Y}_{\text{win}} = \mathbf{A}_{\text{spe}}\mathbf{V}_{\text{spe}} \tag{16}$$

where $\mathbf{Y}_{\text{win}} \in \mathbb{R}^{C\times N^2}$ is the attention output feature.

Then, we perform spatial projection to restore the spatial order

$$\mathbf{Y}'_{\text{spe}} = \mathbf{W}_{\text{win}}\mathbf{Y}_{\text{win}} \tag{17}$$

where $\mathbf{Y}'_{\text{spe}} \in \mathbb{R}^{C\times N^2}$ is the feature with restored spatial order.

We use $\mathbf{Y}_{\text{spe}}^{ij}$ to denote the $\mathbf{Y}'_{\text{spe}}$ with the window indices of i, j. Finally, we reshape and merge outputs

$$\mathbf{Y}_{\text{spe}} = \text{merge}\left(\bigcup_{i=1}^{\lceil H/N\rceil}\bigcup_{j=1}^{\lceil W/N\rceil}\text{reshape}\left(\mathbf{Y}_{\text{spe}}^{ij}\right)\right) \tag{18}$$

where $\mathbf{Y}_{\text{spe}} \in \mathbb{R}^{H\times W\times C}$ is the final output.

Inspired by the conventional window attention [36], after computing the window attention once, we shift the window by a distance of $N/2$ in both the width and height directions, and repeat the SPSA.

IV. Experiments

In this section, we first introduce the experimental setup, including datasets, evaluation metrics, implementation details, and comparison methods. We then conduct quantitative and qualitative comparative experiments against SOTA approaches, followed by a series of ablation studies to validate the effectiveness of the proposed method.

A. Experimental Settings

1) Datasets: We employ three natural hyperspectral datasets, including ARAD_1K [42], ICVL [41], and CAVE [40] and four remote sensing datasets, including Chikusei [43], PaviaC [44], PaviaU [44], and Xiong'an [45].

CAVE [40] contains 32 HSIs, each with a spatial resolution of 512 × 512 pixels and 31 spectral bands spanning the visible range of 400–700 nm at 10-nm intervals. We randomly split the dataset into 22 images for training and ten for testing.

ICVL [41] consists of 201 HSIs, each of size 1392 × 1300 pixels, covering the 400–700-nm spectral range with 31 contiguous bands at 10-nm spacing. After removing redundant scene samples, 100 images are used for training and the remaining 101 for testing.

ARAD_1K [42] includes 950 HSIs of spatial size 482 × 512 pixels, with 31 contiguous spectral bands in the 400–700-nm range. Following the original partitioning, 900 images are allocated for training and 50 for testing.

Chikusei [43] is an airborne hyperspectral dataset acquired over the Chikusei area in Japan using the HH-VNIR-C sensor, covering 128 continuous spectral bands from the visible to the near infrared. Four nonoverlapping 512 × 512 regions are randomly selected as test data, while the rest of the image serves as training data.

PaviaC [44] is a classical urban hyperspectral dataset collected over Pavia, Italy, by the ROSIS airborne sensor, featuring 102 spectral bands. A 256 × 256 region is randomly cropped for testing, with the remainder used for training.

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PaviaU [44] is a subset of PaviaC focusing on the University of Pavia campus, also captured by the ROSIS sensor and containing 103 spectral bands. A 256 × 256 region is randomly selected for testing, and the rest is used for training.

Xiong'an [45] comprises high-resolution airborne HSI over China's Xiong'an New Area, with 256 spectral bands spanning the visible to short-wave infrared range. Three 512 × 512 regions are randomly cropped for testing, while the remaining data is used for training.

RealHSI [46] contains 70 pairs of real-world paired HSI samples for ×4 SR, with each sample comprising 224 spectral bands. We randomly select 10 of these samples for testing, and use the remaining data for training.

2) Metrics: We utilize three widely used metrics to comprehensively evaluate model performance in both spatial and spectral domains: peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) [47] for assessing spatial restoration quality, and spectral angle mapper (SAM) [48] for evaluating spectral fidelity. In addition, we report model parameter count (Params), floating-point operations (FLOPs), and inference time (Time) as supplementary metrics to assess computational efficiency and practical applicability.

3) Implementation Details: Our model is implemented using PyTorch and trained on an NVIDIA RTX 3090 GPU. We use the Adam optimizer with an initial learning rate of 0.001, a batch size of four, and train for 1000 epochs to ensure full convergence. For the loss function, the classical Charbonnier Loss [49] optimizes restoration quality, while spectral angle distance (SAD) [19] constrains spectral consistency.

To simulate the degradation process for generating LR HSIs, we follow the protocol established in [23]. In particular, we blur HR HSIs using a 7 × 7 Gaussian kernel with a standard deviation of 3, followed by spatial downsampling. The downsampling factor depends on the scaling ratio of the superresolution task.

4) Comparison Methods: Multiple methods are compared, including the conventional method bicubic, two convolution-based methods MCNet [24] and Bi3DQRN [23], and six attention-based methods SwinIR [39], SSPSR [19], ESSA [25], HiT-SR [27], SRDNet [26], and VolFormer [21]. All compared methods except bicubic are trained using the same settings as our proposed method.

B. Comparisons With the SOTA Methods

1) Quantitative Comparisons: We compare our proposed methods with existing methods at ×2, ×4, and ×8 superresolution scaling factors. Our proposed framework includes two variants, namely VDSPT (lightweight) and VDSPT-L (large model), which are evaluated alongside existing methods. As illustrated in Table I, which reports results on natural HSI datasets, and Table II, which presents results on remote sensing HSI datasets, VDSPT-L achieves the best performance across all datasets, scaling factors, and evaluation metrics. VDSPT consistently ranks second across nearly all metrics, performing slightly behind VDSPT-L.

2) Qualitative Comparisons: We present superresolution results and corresponding error maps at scale factors of ×2, ×4, and ×8 on the ARAD_1K dataset (see Fig. 6), and similarly on the ICVL dataset (see Fig. 7). In addition, we show superresolution results and error maps at a ×4 scale on the CAVE dataset (see Fig. 8), ×2 superresolution results and error maps for the Chikusei and PaviaC datasets [see Fig. 9(a) and (b)], as well as ×4 superresolution results and error maps for the PaviaU and Xiong'an datasets [see Fig. 9(c) and (d)].

As evident from these visual results, our method consistently produces sharper superresolution images with lower restoration errors across diverse scale factors and datasets of varying sizes. For instance, fine details such as stripes in the ARAD_1K ×2 results and text characters in the CAVE dataset are rendered significantly clearer by our method, while competing methods tend to blur these fine-grained structures and produce noticeable artifacts. This performance gap is rooted in the inherent limitations of conventional attention mechanisms, which our proposed VDSA and SPSA modules are designed to address: VDSA enhances the efficiency of global spatial feature extraction, while SPSA ensures more stable and accurate spectral feature modeling.

Furthermore, Fig. 10 shows the comparison of the restored spectral curves under ×4 superresolution on the ICVL dataset, Fig. 11 presents the corresponding results for the CAVE dataset at ×4, and Fig. 12 shows those for the Chikusei dataset at ×2. The results consistently demonstrate that the proposed VDSPT model generates spectral curves closely aligned with the ground truth (GT) and achieves lower per-channel errors, thereby confirming the superior performance of the VDSA and SPSA modules.

Fig. 13 presents ×4 SR visual results on real-world LR HSIs. Competing methods suffer from oversmoothing or edge blurring, while VDSPT achieves sharper edges. The red-boxed details verify its superior boundary recovery, demonstrating the effectiveness and practicality of our method.

3) Computational Efficiency Analysis: At the ×2 scaling factor, targeting the restoration of 512 × 512 × 31 HR HSI, we compare the model parameters and computational complexity of each method (results are presented in Table III). Experiments demonstrate that the proposed method has a parameter count comparable to existing solutions but significantly reduces computational complexity. For instance, compared to VolFormer, the SOTA competing method in terms of superresolution performance, VDSPT requires only 2% of its computational cost while maintaining comparable performance. VDSPT-L requires just 7% of its computational cost while achieving substantial performance improvements. This indicates that our method achieves notable improvements in both restoration accuracy and computational efficiency, providing greater feasibility for deployment on edge devices.

We conduct a module-level computational efficiency ablation analysis on the two core modules, VDSA and SPSA. The results (Table IV) show distinct computational characteristics of the two modules. SPSA dominates the model's parameters and FLOPs overhead, while VDSA contributes more to inference latency.

C. Discussion

1) Ablation on Key Components: We conduct ablation experiments at ×4 scale on the ARAD_1K dataset. As

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TABLE I

QUANTITATIVE COMPARISON ON NATURAL HSI DATASETS ARAD_1K, ICVL, AND CAVE USING SUPERRESOLUTION SCALE FACTORS OF 2, 4, AND 8. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	Method	×2			×4			×8		
		PSNR↑	SSIM↑	SAM↓	PSNR↑	SSIM↑	SAM↓	PSNR↑	SSIM↑	SAM↓
CAVE [41]	Bicubic	33.87	0.933	4.978	33.40	0.925	5.245	30.11	0.869	6.932
	SSPSR [19]	42.52	0.980	4.116	38.20	0.957	4.819	32.31	0.908	6.043
	MCNet [25]	43.67	0.986	4.366	37.26	0.956	4.975	32.18	0.906	5.721
	Bi3DQRN [24]	42.05	0.985	4.200	37.23	0.953	4.973	32.05	0.906	6.022
	SwinIR [40]	42.16	0.986	4.113	37.25	0.954	4.276	32.41	0.911	6.173
	ESSA [26]	41.80	0.979	4.279	37.13	0.950	5.787	31.41	0.887	6.526
	HiT-SR [28]	41.63	0.973	4.514	37.02	0.949	6.101	31.24	0.872	6.731
	SRDNet [27]	43.85	0.983	4.012	38.46	0.959	4.823	32.92	0.908	5.834
	VolFormer [21]	44.19	0.986	3.963	38.81	0.960	4.676	33.04	0.917	5.515
	VDSPT (Ours)	45.12	0.987	3.828	39.65	0.965	4.331	33.46	0.924	5.312
	VDSPT-L (Ours)	**45.83**	**0.989**	**3.788**	**40.18**	**0.971**	**4.249**	**33.79**	**0.927**	**5.258**
ICVL [42]	Bicubic	41.77	0.959	1.548	41.11	0.952	1.631	37.37	0.908	2.280
	SSPSR [19]	48.93	0.994	1.284	42.28	0.959	1.765	37.85	0.918	2.205
	MCNet [25]	50.60	0.994	0.908	42.32	0.960	1.717	37.66	0.918	2.175
	Bi3DQRN [24]	49.89	0.994	1.021	41.12	0.952	1.655	38.02	0.919	2.197
	SwinIR [40]	50.49	0.993	1.001	42.27	0.958	1.826	37.77	0.919	2.185
	ESSA [26]	49.56	0.993	1.088	42.31	0.959	1.775	37.82	0.916	2.265
	HiT-SR [28]	48.69	0.992	1.205	42.06	0.952	1.822	37.84	0.918	2.173
	SRDNet [27]	50.69	0.995	0.999	42.26	0.959	1.765	38.17	0.920	2.189
	VolFormer [21]	50.67	0.995	1.025	44.35	0.972	1.473	38.05	0.921	2.193
	VDSPT (Ours)	50.81	0.996	0.972	44.13	0.970	1.451	38.14	0.921	2.172
	VDSPT-L (Ours)	**51.70**	**0.996**	**0.881**	**44.43**	**0.973**	**1.430**	**38.24**	**0.922**	**2.137**
ARAD_1K [43]	Bicubic	36.57	0.908	1.727	36.11	0.898	1.867	33.37	0.845	2.834
	SSPSR [19]	42.64	0.970	1.404	37.32	0.917	2.467	33.43	0.846	2.993
	MCNet [25]	44.74	0.980	1.004	38.07	0.925	1.585	33.67	0.847	2.653
	Bi3DQRN [24]	44.71	0.980	1.096	39.28	0.939	1.744	34.34	0.869	2.499
	SwinIR [40]	44.62	0.979	1.423	38.24	0.928	1.674	33.68	0.847	2.627
	ESSA [26]	43.88	0.978	1.265	37.81	0.922	1.893	33.43	0.847	2.850
	HiT-SR [28]	43.89	0.978	1.187	38.26	0.929	1.687	33.86	0.849	2.668
	SRDNet [27]	45.87	0.983	1.165	39.98	0.945	1.541	34.43	0.871	2.565
	VolFormer [21]	47.13	0.988	0.924	40.82	0.952	1.362	34.70	0.877	2.253
	VDSPT (Ours)	48.34	0.986	0.887	41.15	0.955	1.316	35.13	0.882	2.243
	VDSPT-L (Ours)	**49.15**	**0.992**	**0.710**	**41.71**	**0.959**	**1.207**	**35.55**	**0.890**	**2.102**

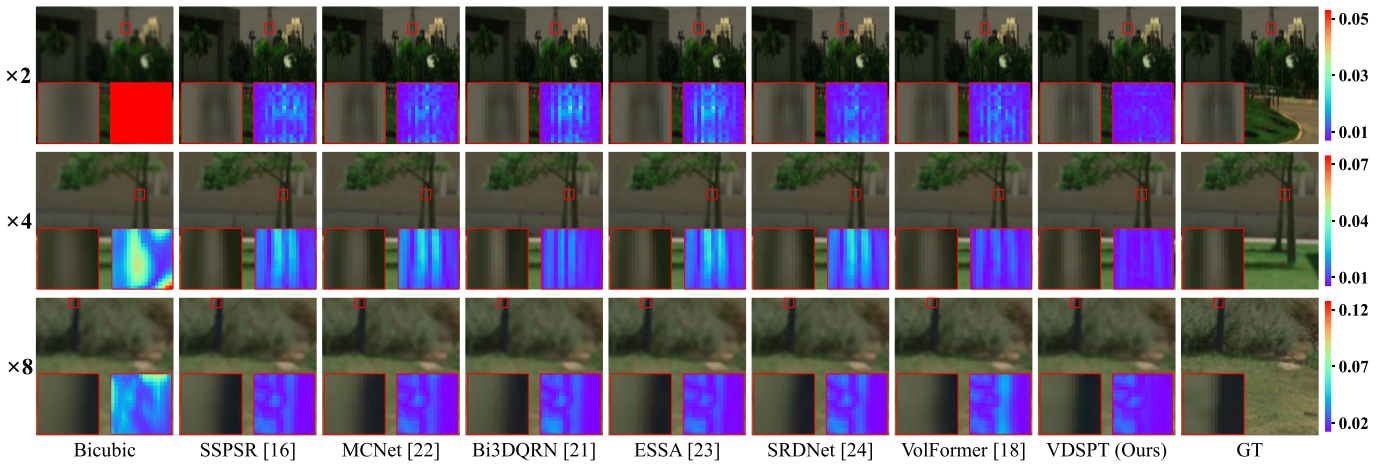


Fig. 7. Visual comparison on ICVL [41] at ×2, ×4, ×8 scales. We use band numbers (27, 14, 6) to generate the pseudocolor images. (Bottom-left) for each result, selected details are highlighted with red boxes, and (bottom-right) the corresponding superresolution results and error maps are enlarged.

illustrated in Table V, compared to conventional spatial attention and spectral attention, the proposed VDSA and SPSA both deliver significant performance improvements. Moreover, their combination yields even greater performance improvements.

2) Ablation on the Differential Mechanism in VDSA: To validate that the subtraction-based differential operation is the key contributor to VDSA's performance (rather than merely increased model capacity), we design comparative experiments against two baseline variants: 1) a single-branch attention module that retains only one spatial attention map without the differential operation and 2) alternative fusion strategies replacing subtraction (i.e., elementwise addition, weighted summation, and concatenation) while keeping other module configurations identical. As illustrated in Table VI, the proposed subtraction-based differential operation achieves the best performance among all settings. All alternative variants, including the single-branch baseline and other fusion strategies, lead to obvious performance degradation. These results

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TABLE II

QUANTITATIVE COMPARISON ON REMOTE SENSING HSI DATASETS CHIKUSEI, PAVIAC, PAVIAU, AND XIONG'AN USING SUPERRESOLUTION SCALE FACTORS OF 2, 4, AND 8. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	Method	×2			×4			×8		
		PSNR↑	SSIM↑	SAM↓	PSNR↑	SSIM↑	SAM↓	PSNR↑	SSIM↑	SAM↓
Chikusei [44]	Bicubic	37.76	0.901	3.808	36.36	0.873	5.038	34.16	0.826	7.417
	SSPSR [19]	42.99	0.970	2.089	38.05	0.907	3.889	35.04	0.845	5.551
	MCNet [25]	42.68	0.968	2.158	36.36	0.885	4.342	33.43	0.819	6.554
	Bi3DQRN [24]	37.81	0.903	3.938	37.36	0.893	4.037	35.16	0.846	5.417
	SwinIR [40]	42.55	0.963	2.252	37.64	0.898	3.935	34.11	0.837	6.146
	ESSA [26]	41.31	0.958	2.535	35.60	0.872	4.902	33.08	0.809	7.440
	HiT-SR [28]	42.25	0.961	2.413	36.89	0.886	4.149	33.99	0.8260	6.237
	SRDNet [27]	43.69	0.975	2.002	38.21	0.908	3.787	35.15	0.846	5.440
	VolFormer [21]	44.08	0.977	2.082	39.59	0.932	3.209	35.40	0.857	5.095
	VDSPT (Ours)	45.87	0.982	1.772	40.33	0.941	2.723	35.51	0.862	4.732
	VDSPT-L (Ours)	**46.57**	**0.987**	**1.384**	**40.61**	**0.946**	**2.490**	**35.67**	**0.866**	**4.463**
PaviaC [45]	Bicubic	27.19	0.675	7.078	25.84	0.609	8.273	24.54	0.487	9.641
	SSPSR [19]	31.66	0.881	5.715	27.68	0.712	7.621	24.81	0.511	8.927
	MCNet [25]	31.14	0.869	5.513	25.97	0.638	7.405	23.36	0.436	9.307
	Bi3DQRN [24]	27.69	0.715	7.405	26.84	0.649	7.277	24.84	0.510	8.642
	SwinIR [40]	30.82	0.864	5.775	26.55	0.667	7.821	24.76	0.508	8.723
	ESSA [26]	30.05	0.838	6.040	25.19	0.589	8.355	23.17	0.426	11.18
	HiT-SR [28]	30.57	0.852	6.113	26.22	0.621	7.834	24.24	0.504	8.983
	SRDNet [27]	32.09	0.889	5.718	27.68	0.712	7.143	24.84	0.510	8.657
	VolFormer [21]	32.27	0.891	5.983	28.52	0.761	7.744	24.95	0.533	8.730
	VDSPT (Ours)	33.47	0.914	5.124	28.92	0.782	7.101	24.95	0.538	8.433
	VDSPT-L (Ours)	**34.51**	**0.931**	**4.685**	**29.14**	**0.794**	**6.207**	**24.96**	**0.544**	**8.168**
PaviaU [45]	Bicubic	27.55	0.724	5.597	27.23	0.706	5.791	25.10	0.576	8.326
	SSPSR [19]	31.78	0.886	4.086	27.89	0.742	5.669	25.03	0.606	7.507
	MCNet [25]	31.27	0.878	4.073	25.90	0.696	6.179	22.90	0.550	8.793
	Bi3DQRN [24]	27.84	0.740	5.692	27.23	0.706	5.790	25.10	0.606	7.326
	SwinIR [40]	31.82	0.885	4.201	27.91	0.742	6.347	25.11	0.613	7.232
	ESSA [26]	29.86	0.840	4.716	24.93	0.654	7.225	22.88	0.542	9.480
	HiT-SR [28]	31.49	0.873	4.864	27.45	0.729	6.624	24.99	0.604	7.538
	SRDNet [27]	32.08	0.889	4.238	27.94	0.742	5.509	25.09	0.606	7.334
	VolFormer [21]	31.96	0.892	4.315	28.70	0.779	5.236	25.21	0.620	7.203
	VDSPT (Ours)	33.57	0.916	4.012	29.29	0.801	4.972	25.32	0.629	6.996
	VDSPT-L (Ours)	**34.57**	**0.937**	**3.351**	**29.57**	**0.813**	**4.622**	**25.38**	**0.638**	**6.782**
Xiong'an [46]	Bicubic	45.35	0.967	0.720	43.07	0.955	0.960	41.37	0.949	1.675
	SSPSR [19]	48.01	0.985	0.625	45.39	0.969	0.811	42.76	0.954	1.360
	MCNet [25]	47.88	0.985	0.574	44.07	0.963	0.793	41.42	0.947	1.034
	Bi3DQRN [24]	45.63	0.968	0.741	45.06	0.965	0.740	43.37	0.954	0.875
	SwinIR [40]	48.25	0.986	0.593	45.42	0.969	0.807	42.82	0.954	1.082
	ESSA [26]	47.03	0.981	0.690	43.36	0.960	0.962	40.75	0.946	1.813
	HiT-SR [28]	48.01	0.985	0.613	45.37	0.967	0.822	42.13	0.952	1.287
	SRDNet [27]	48.59	0.987	0.569	45.72	0.969	0.720	43.34	0.954	0.885
	VolFormer [21]	48.83	0.988	0.601	46.03	0.972	0.723	43.35	0.954	0.901
	VDSPT (Ours)	49.32	0.989	0.551	46.23	0.973	0.682	43.45	0.955	0.873
	VDSPT-L (Ours)	**49.73**	**0.990**	**0.520**	**46.49**	**0.974**	**0.665**	**43.59**	**0.956**	**0.866**

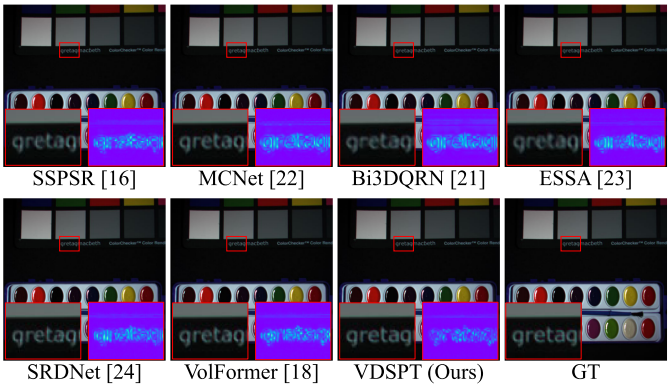


Fig. 8. Visual comparison on CAVE [40] at ×4 scale. (Bottom-left) for each result, selected details are highlighted with red boxes, and (bottom-right) the corresponding superresolution results and error maps are enlarged.

TABLE III

COMPARISON WITH EXISTING ATTENTION-BASED SINGLE HSI-SR METHODS IN TERMS OF THE NUMBER OF PARAMETERS, COMPUTATIONAL COST, AND INFERENCE TIME FOR ×2 SUPERRESOLUTION, TARGETING AN OUTPUT RESOLUTION OF 512 × 512 × 31

Method	ESSA [26]	SRDNet [27]	VolFormer [21]	VDSPT	VDSPT-L
Params (M)	8.69	**1.53**	23.01	3.00	8.40
FLOPs (G)	3333.47	2845.78	7042.72	**140.27**	491.85
Time (ms)	2.33	2.97	91.71	**1.63**	2.62

TABLE IV

MODULE-LEVEL COMPUTATIONAL EFFICIENCY BREAKDOWN OF VDSPT FOR ×2 SUPERRESOLUTION, TARGETING AN OUTPUT RESOLUTION OF 512 × 512 × 31

Setting	Params (M) ↓	FLOPs (G) ↓	Time (ms) ↓
Full VDSPT model	3.00	140.27	1.63
Model without VDSA	2.78	139.86	0.89
Model without SPSA	0.45	12.34	1.02

verify that the performance gain of VDSA comes from the unique differential mechanism, rather than the increased model capacity.

3) Ablation on the Residual Connection in VDSA: To verify the necessity of the residual connection in VDSA (claimed to

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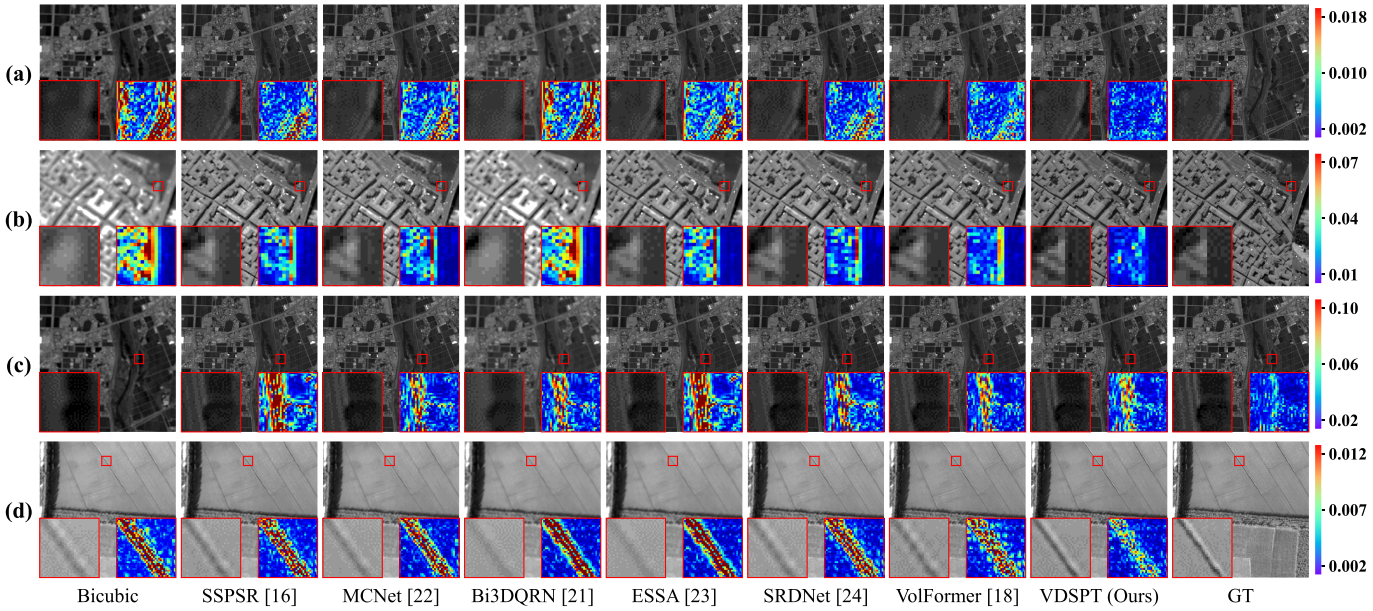


Fig. 9. Visual comparison on remote sensing datasets. (Bottom-left) for each result, selected details are highlighted with red boxes, and (bottom-right) the corresponding superresolution results and error maps are enlarged. (a) Comparison of $\times 2$ superresolution results on Chikusei [43]. (b) Comparison of $\times 2$ superresolution results on PaviaC [44]. (c) Comparison of $\times 4$ superresolution results on PaviaU [44]. (d) Comparison of $\times 4$ superresolution results on Xiong'an [45].

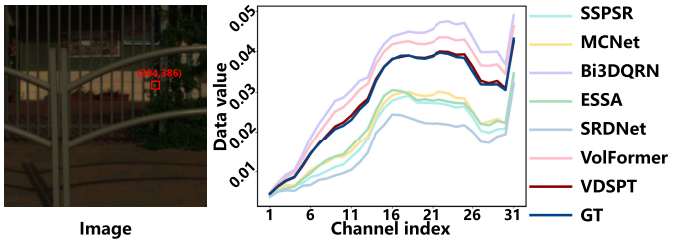


Fig. 10. Spectral comparison on ICVL [41] at $\times 4$ scale. (Right) The spectral curves show the average spectral responses of different methods and (left) the GT at the spatial location corresponding to the red box.

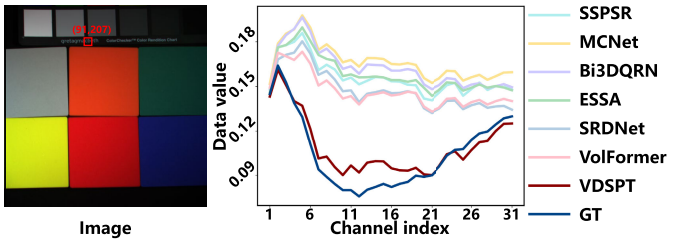


Fig. 11. Spectral comparison on CAVE [40] at $\times 4$ scale. (Right) The spectral curves show the average spectral responses of different methods and (left) the GT at the spatial location corresponding to the red box.

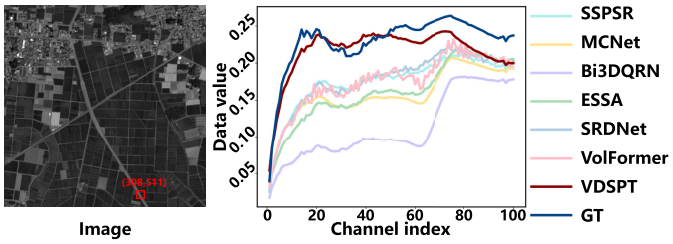


Fig. 12. Spectral comparison on Chikusei [43] at $\times 2$ scale. (Right) The spectral curves show the average spectral responses of different methods and (left) the GT at the spatial location corresponding to the red box.

TABLE V

BREAKDOWN ABLATION OF THE PROPOSED VDSPT ON ARAD_1K [42] AT $\times 2$ SCALE. SPA ATTN DENOTES THE SPATIAL ATTENTION TYPE, SPEC ATTN DENOTES THE SPECTRAL ATTENTION TYPE, NORMAL REPRESENTS CONVENTIONAL SPATIAL ATTENTION AND SPECTRAL ATTENTION

Spa Attn	Spec Attn	PSNR↑	SSIM↑	SAM↓
Normal	Normal	39.87	0.941	1.589
VDSA	Normal	40.77	0.951	1.391
Normal	SPSA	40.83	0.952	1.344
VDSA	SPSA	**41.15**	**0.955**	**1.316**

stabilize the training process), we conduct a dedicated ablation study by removing the residual terms in (3)–(5) and comparing the resulting model's performance with the original VDSA. As illustrated in Table VII, removing the residual connection leads to an obvious performance drop in all evaluation metrics. These results verify that the residual connection effectively stabilizes the training process and improves the final reconstruction performance of the model.

4) Ablation on Window Shifting in SPSA: To explicitly confirm the contribution of window shifting in SPSA (adopted to enhance cross-window spatial dependencies), we conduct a dedicated ablation study by removing the window shifting operation and comparing the resulting model's performance with the original SPSA. As illustrated in Table VIII, removing window shifting leads to a significant performance degradation across all metrics. These results demonstrate that window shifting effectively improves the spatial consistency and

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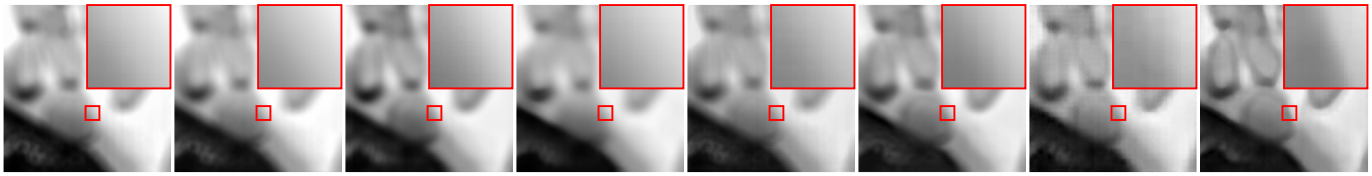


Fig. 13. Visual comparison of ×4 SR results on the real-world RealHSI dataset. For each result, selected details are highlighted with red boxes, and the corresponding enlarged views are shown at the bottom of each image.

TABLE VI

ABLATION STUDY ON THE DIFFERENTIAL MECHANISM IN VDSA ON THE ARAD_1K DATASET FOR ×4 SR. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Setting	PSNR (dB) ↑	SSIM ↑	SAM ↓
Differential operation (Ours)	**41.71**	**0.959**	**1.207**
Element-wise addition	40.84	0.949	1.679
Weighted summation	<u>41.13</u>	<u>0.954</u>	<u>1.316</u>
Concatenation	40.95	0.953	1.322
Single-branch attention	40.62	0.950	1.443

TABLE VII

ABLATION STUDY ON THE RESIDUAL CONNECTION IN VDSA ON THE ARAD_1K DATASET FOR ×4 SUPERRESOLUTION. THE BEST RESULTS ARE SHOWN IN BOLD

Setting	PSNR (dB) ↑	SSIM ↑	SAM ↓
With residual connection	**41.71**	**0.959**	**1.207**
Without residual connection	41.12	0.951	1.578

TABLE VIII

ABLATION STUDY ON WINDOW SHIFTING IN SPSA ON THE ARAD_1K DATASET FOR ×4 SUPERRESOLUTION. THE BEST RESULTS ARE SHOWN IN BOLD

Setting	PSNR (dB) ↑	SSIM ↑	SAM ↓
With window shifting	**41.71**	**0.959**	**1.207**
Without window shifting	40.02	0.943	1.873

TABLE IX

CONTROL EXPERIMENT ON CHANNEL ORDER PRESERVATION IN SPSA ON THE ARAD_1K DATASET FOR ×4 SUPERRESOLUTION. THE BEST RESULTS ARE SHOWN IN BOLD

Setting	PSNR (dB) ↑	SSIM ↑	SAM ↓
Without channel shuffle	**41.71**	**0.959**	**1.207**
With channel shuffle	41.55	0.957	1.246

spectral fidelity of the superresolution results, which is a critical component of our SPSA module.

5) Validation of Channel Order Preservation in SPSA: To verify that preserving channel order via spatial projection in SPSA is critical for spectral consistency, we design a control experiment: we randomly shuffle the channel order of the input feature map before spectral attention calculation in SPSA (and restore the original order after calculation) while keeping all other settings unchanged. As illustrated in Table IX, when the channel order is shuffled, the model exhibits an obvious performance drop, especially in the SAM metric that directly reflects spectral fidelity. These results verify that preserving the original channel order via spatial projection in SPSA is critical for maintaining spectral consistency and overall superresolution performance.

6) Robustness Evaluation Under Noisy Inputs: We compare our proposed VDSPT with existing methods via ×4 superresolution experiments on the ARAD_1K dataset, under three levels of synthetic noise covering common real-world HSI corruptions.

1) *Easy*: Gaussian noise ($\sigma = 0.05$).
2) *Medium*: Poisson noise (peak = 500.0) + Gaussian noise ($\sigma = 0.05$).
3) *Hard*: Poisson noise (peak = 800.0) + Gaussian noise ($\sigma = 0.05$) + vertical stripe noise (stripe_prob = 0.2, intensity = 0.2).

The proposed VDSPT is evaluated alongside existing methods under identical experimental settings. As illustrated in Table X, VDSPT achieves the best performance across all noise levels and evaluation metrics, consistently outperforming all competing methods. These results demonstrate that VDSPT exhibits strong robustness to various noise corruptions.

7) Window Size Analysis: We analyze the influence of window size at ×4 scale on the ARAD_1K dataset. As illustrated in Table XI, the model achieves optimal performance with a window size of 16. At this scale, windows naturally encompass local objects with similar characteristics while retaining sufficient spatial information. Conversely, smaller window sizes (4 and 8) suffer from insufficient spatial information, while larger windows (size 32) become vulnerable to spectral heterogeneity interference when containing multiple distinct objects.

8) Model Size Analysis: We analyze the influence of model size at ×4 scale on the ARAD_1K dataset. As illustrated in Table XII, VDSPT exhibits progressively improved performance with increasing network depth and feature dimensions. This demonstrates that VDSPT is not constrained by lightweight design principles and possesses significant potential for performance enhancement when scaling up model parameters.

9) Spatial Feature Extraction Efficiency: We remove the SPSA module from the original model and retain only the VDSA module, comparing it against a variant in which the VDSA is replaced with a conventional spatial attention module. All experiments are conducted on the ARAD_1K dataset for ×4 superresolution. We fix the window size to eight, the feature dimension to 32, and the number of stages to two while gradually increasing the total number of modules across all stages from 5 to 15 to evaluate the restoration performance of both configurations. As illustrated in Fig. 14, the VDSA-based model exhibits a clear trend of performance improvement

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TABLE X

QUANTITATIVE COMPARISON OF ×4 SUPERRESOLUTION UNDER DIFFERENT NOISE LEVELS ON THE ARAD_1K DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method	Easy Level			Medium Level			Hard Level		
	PSNR ↑	SSIM ↑	SAM ↓	PSNR ↑	SSIM ↑	SAM ↓	PSNR ↑	SSIM ↑	SAM ↓
SSPSR [19]	36.21	0.894	3.245	35.62	0.886	3.767	34.88	0.884	4.484
MCNet [25]	36.22	0.902	3.034	35.58	0.885	4.124	35.03	0.884	4.797
Bi-3DQRN [24]	35.56	0.885	4.017	34.82	0.874	5.213	33.76	0.864	6.638
ESSA [26]	35.85	0.892	3.627	35.20	0.883	3.815	34.44	0.876	4.811
SRDNet [27]	36.37	0.899	2.396	35.72	0.888	3.591	35.06	0.885	4.420
VolFormer [21]	37.01	0.915	2.238	36.37	0.899	3.191	35.11	0.885	4.337
VDSPT (Ours)	**37.23**	**0.921**	**1.914**	**36.46**	**0.903**	**2.971**	**35.17**	**0.886**	**4.109**

TABLE XI

ABLATION STUDY OF WINDOW SIZES ON ARAD_1K [42] AT ×2 SCALE

Window Size	PSNR↑	SSIM↑	SAM↓
4	40.11	0.941	1.491
8	40.92	0.953	1.342
16	**41.15**	**0.955**	**1.316**
32	41.03	0.953	1.325

TABLE XII

PERFORMANCE COMPARISON ACROSS DIFFERENT MODEL SIZES ON ARAD_1K [42] AT ×2 SCALE. DIM DENOTES FEATURE CHANNEL DIMENSIONS AND STAGE REPRESENTS THE NUMBER OF DOWNSAMPLING–UPSAMPLING STAGES

Dim	Stage	PSNR↑	SSIM↑	SAM↓	Params (M)	FLOPs (G)
16	1	40.12	0.947	1.513	2.74	58.19
16	2	40.54	0.949	1.474	4.01	70.62
32	1	41.15	0.955	1.316	3.00	140.27
32	2	41.36	0.956	1.281	4.91	176.13
64	1	41.54	0.958	1.240	3.98	376.09
64	2	**41.71**	**0.959**	**1.207**	8.40	491.85

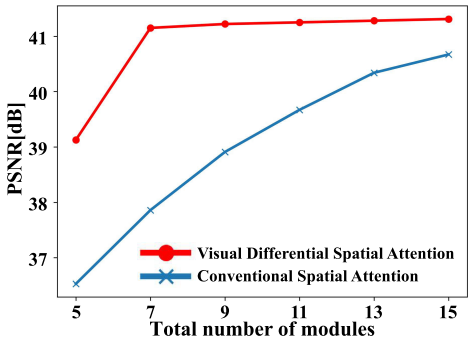


Fig. 14. Comparison of the impact of varying the number of modules in VDSA and conventional spatial attention on superresolution performance on ARAD_1K [42] at ×2 scale.

followed by saturation. It achieves excellent restoration quality with just seven modules. On the contrary, the model employing conventional spatial attention performs poorly with fewer modules and only reaches comparable performance when the module count is increased to 12. Furthermore, across all tested settings, the VDSA-based model consistently outperforms the one using conventional spatial attention. These results demonstrate that VDSA enables highly efficient spatial feature extraction, delivering strong performance with significantly fewer stacked modules and thereby highlighting its superior modeling capacity and computational efficiency.

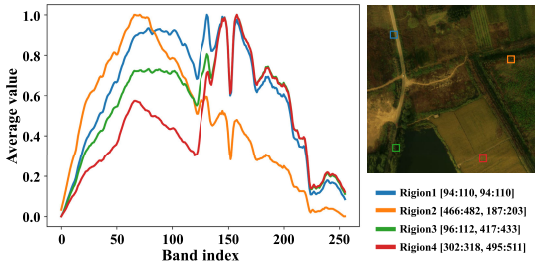


Fig. 15. Comparison of spectral curves across different regions in Xiong'an [45]. The different colored curves represent the average spectral reflectance values of the HSI within the corresponding color-bounded regions.

10) Cross-Regional Spectral Curve Comparison: We present the spectral curves from different regions of the same image, as illustrated in Fig. 15. These curves can exhibit substantial variations even within a single image. Global spectral attention mechanisms, which apply the same attention weights uniformly across the entire image, fail to capture such local discrepancies. On the contrary, our SPSA computes spectral attention within local windows, enabling it to adaptively and more precisely extract region-specific spectral features.

V. CONCLUSION

This article proposes VDSPT, an efficient lightweight framework for HSI-SR. It introduces VDSA, which calibrates overconcentrated local weight bias of conventional attention via subtraction between two spatial attention maps, improving global spatial feature extraction efficiency. We further design SPSA, whose spatial-dimension projection addresses the channel confusion problem in existing spectral attention methods. Benefiting from the two core modules, VDSPT achieves competitive performance with a lightweight cascaded structure, without deep network architectures. Experimental results show that VDSPT outperforms SOTA methods in both reconstruction quality and computational efficiency.

For future work, we will extend the proposed framework to other HSI restoration tasks and explore how the restored results can better benefit downstream applications.

REFERENCES

[1] G. ElMasry and D.-W. Sun, "Principles of hyperspectral imaging technology," in *Hyperspectral Imaging for Food Quality Analysis and Control*. San Diego, CA, USA: Academic, 2010, pp. 3–43.

[2] A. Plaza et al., "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Jul. 2009.

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[3] R. A. Schultz, T. Nielsen, J. R. Zavaleta, R. Ruch, R. Wyatt, and H. R. Garner, "Hyperspectral imaging: A novel approach for microscopic analysis," *Cytometry*, vol. 43, no. 4, pp. 239–247, 2001.

[4] A. Munawar, R. Hayati, and F. Fachruddin, "Rapid determination of inner quality parameters of intact mango fruits using portable near infrared spectroscopy," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 711, no. 1, 2021, Art. no. 012009.

[5] M. L. Clark, J. Buck-Diaz, and J. Evens, "Mapping of forest alliances with simulated multi-seasonal hyperspectral satellite imagery," *Remote Sens. Environ.*, vol. 210, pp. 490–507, Jun. 2018.

[6] R. M. Levenson and J. R. Mansfield, "Multispectral imaging in biology and medicine: Slices of life," *Cytometry A*, vol. 69A, no. 8, pp. 748–758, Aug. 2006.

[7] G. Lu and B. Fei, "Medical hyperspectral imaging: A review," *J. Biomed. Opt.*, vol. 19, no. 1, Jan. 2014, Art. no. 010901.

[8] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[9] A. Chakrabarti and T. Zickler, "Statistics of real-world hyperspectral images," in *Proc. CVPR*, Jun. 2011, pp. 193–200.

[10] Y. Yu, E. Pan, X. Wang, Y. Wu, X. Mei, and J. Ma, "Unmixing before fusion: A generalized paradigm for multi-source-based hyperspectral image synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 9297–9306.

[11] Z. Lai, Y. Fu, and J. Zhang, "Hyperspectral image super resolution with real unaligned RGB guidance," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 2999–3011, Feb. 2025.

[12] Y. Zhang, Z. Lai, T. Zhang, Y. Fu, and C. Zhou, "Unaligned RGB guided hyperspectral image super-resolution with spatial–spectral concordance," *Int. J. Comput. Vis.*, vol. 133, no. 9, pp. 6590–6610, Sep. 2025.

[13] Y. Fu, T. Zhang, L. Wang, and H. Huang, "Coded hyperspectral image reconstruction using deep external and internal learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3404–3420, Jul. 2022.

[14] H. Li and Y. Fu, "FCDFusion: A fast, low color deviation method for fusing visible and infrared image pairs," *Comput. Vis. Media*, vol. 11, no. 1, pp. 195–211, Feb. 2025.

[15] Y. Liu, R. Dian, and S. Li, "Low-rank transformer for high-resolution hyperspectral computational imaging," *Int. J. Comput. Vis.*, vol. 133, no. 2, pp. 809–824, Feb. 2025.

[16] R. Dian, A. Guo, and S. Li, "Zero-shot hyperspectral sharpening," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12650–12666, Oct. 2023.

[17] R. Dian, T. Shan, W. He, and H. Liu, "Spectral super-resolution via model-guided cross-fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 10059–10070, Jul. 2024.

[18] S. Mei, X. Yuan, J. Ji, Y. Zhang, S. Wan, and Q. Du, "Hyperspectral image spatial super-resolution via 3D full convolutional neural network," *Remote Sens.*, vol. 9, no. 11, p. 1139, 2017.

[19] J. Jiang, H. Sun, X. Liu, and J. Ma, "Learning spatial–spectral prior for super-resolution of hyperspectral imagery," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1082–1096, 2020.

[20] X. Su et al., "Eigensr: Eigenimage-bridged pre-trained rgb learners for single hyperspectral image super-resolution," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 7, pp. 7033–7041.

[21] D. Yu and Z. Gao, "Volformer: Explore more comprehensive cube interaction for hyperspectral image restoration and beyond," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2025, pp. 28091–28101.

[22] M. Li et al., "Latent diffusion enhanced rectangle transformer for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 1, pp. 549–564, Jan. 2024.

[23] Y. Fu, Z. Liang, and S. You, "Bidirectional 3D quasi-recurrent neural network for hyperspectral image super-resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2674–2688, 2021.

[24] Q. Li, Q. Wang, and X. Li, "Mixed 2D/3D convolutional network for hyperspectral image super-resolution," *Remote Sens.*, vol. 12, no. 10, p. 1660, May 2020.

[25] M. Zhang, C. Zhang, Q. Zhang, J. Guo, X. Gao, and J. Zhang, "ESSAformer: Efficient transformer for hyperspectral image super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 23016–23027.

[26] T. Liu, Y. Liu, C. Zhang, L. Yuan, X. Sui, and Q. Chen, "Hyperspectral image super-resolution via dual-domain network based on hybrid convolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512518.

[27] X. Zhang, Y. Zhang, and F. Yu, "Hit-SR: Hierarchical transformer for efficient image super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 483–500.

[28] Z. Wu, Y. Chen, N. Yokoya, and W. He, "MP-HSIR: A multi-prompt framework for universal hyperspectral image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 13009–13020.

[29] B. Wang, D. Wang, H. Guo, Y. Fu, and J. Zhang, "Degradation-aware metric prompting for hyperspectral image restoration," 2025, *arXiv:2512.20251*.

[30] W. Dong, L. Zhang, G. Shi, and X. Wu, "Nonlocal back-projection for adaptive image enlargement," in *Proc. IEEE Int. Conf. Image Process.*, Mar. 2009, pp. 349–352.

[31] J. Yang, J. Wright, T. S. Huang, and Y. Ma, "Image super-resolution via sparse representation," *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[32] Q. Wang, Q. Li, and X. Li, "Hyperspectral image superresolution using spectrum and feature context," *IEEE Trans. Ind. Electron.*, vol. 68, no. 11, pp. 11276–11285, Nov. 2021.

[33] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–18.

[34] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[35] Y. Tay, M. Dehghani, D. Bahri, and D. Metzler, "Efficient transformers: A survey," *ACM Comput. Surveys*, vol. 55, no. 6, pp. 1–28, 2022.

[36] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[37] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5718–5729.

[38] X. Liu, H. Peng, N. Zheng, Y. Yang, H. Hu, and Y. Yuan, "EfficientViT: Memory efficient vision transformer with cascaded group attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14420–14430.

[39] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.

[40] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[41] B. Arad and O. Ben-Shahar, "Sparse recovery of hyperspectral signal from natural RGB images," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 19–34.

[42] B. Arad et al., "NTIRE 2022 spectral recovery challenge and data set," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 863–881.

[43] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Dept. Space Appl. Lab., Univ. Tokyo, Tokyo, Japan, Tech. Rep. SAL-2016-05-27, 2016, vol. 5, no. 5, p. 5.

[44] X. Huang and L. Zhang, "A comparative study of spatial approaches for urban mapping using hyperspectral ROSIS images over Pavia city, northern Italy," *Int. J. Remote Sens.*, vol. 30, no. 12, pp. 3205–3221, Jun. 2009.

[45] C. Yi et al., "Aerial hyperspectral remote sensing classification dataset of Xiongan new area (Matiwan village)," *Nat. Remote Sens. Bull.*, vol. 24, no. 11, pp. 1299–1306, 2020.

[46] A. Ulrichsen, T. De Kerf, R. D. Dunphy, P. Murray, S. Vanlanduit, and S. Marshall, "A true hyperspectral image super-resolution dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2025, pp. 4412–4421.

[47] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[48] R. H. Yuhas, A. F. Goetz, and J. W. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (sam) algorithm," in *Proc. JPL, Summaries 3rd Annu. JPL Airborne Geosci. Workshop*, vol. 1, 1992, pp. 1–18.

[49] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Deep Laplacian pyramid networks for fast and accurate super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5835–5843.

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